Hamza El Byad, Ph.D.

Statistical Time-Series Researcher | Machine Learning | High-Energy Astrophysics

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Research Profile

Ph.D. in Mathematics & Computer Science, specializing in statistical time-series analysis, statistical inference and high-energy astrophysics. First-author peer-reviewed research in X-ray timing; additional experience in industrial machine learning, scientific data analysis and applied AI.

Research & Professional Experience

University of Cagliari **2022 – 2025**
Doctoral Researcher | Cagliari, Italy

- Analyzed 39 NuSTAR observations (2014–2024) to investigate X-ray variability and QPO evolution in M82 X-1 across 20–300 mHz.
- Applied Fourier-domain power-spectral analysis, significance assessment and uncertainty-aware statistical inference to noisy observations.
- Explored regularized inverse methods and resampling techniques for time-series reconstruction and uncertainty estimation.
- Led first-author research published in The Astrophysical Journal, 989(2), 202 (2025).

Rombo AI **Sep 2025 – Aug 2026**
Machine Learning Engineer | Cagliari, Italy

- Developed NMR spectroscopy preprocessing and chemometric regression workflows for industrial property prediction, with grouped validation across repeated measurements.
- Worked on predictive models for process monitoring; contributed to an internal RAG application for catalog retrieval and traceable estimates.
- Developed a traffic-signal simulation and control framework using SUMO/TraCI to compare fixed-time, Max Pressure and reinforcement-learning approaches.

European Space Agency (ESAC) **Jan – Jul 2023**
Data Science Research Intern | Madrid, Spain

- Applied deep-learning methods to compact-source detection in Planck sky maps and evaluated astronomical imaging data.

Astronomical Observatory of Cagliari (INAF-OAC) **Dec 2020 – Jul 2021**
Research Intern | Cagliari, Italy

- Developed Python-based analysis of X-ray time series and statistical investigations of quasiperiodic oscillations.

Additional Quantitative Research

Stochastic Inventory Control (B.Sc. Thesis)
Modeled an inventory system with reusable returns as a continuous-time Markov Decision Process and established base-stock policy optimality.

Research & Technical Skills

Statistical Research	Statistical inference, Fourier and power-spectral analysis, time-series modeling, significance assessment, bootstrap methods
Astrophysics	High-energy and X-ray astrophysics, QPO variability, NuSTAR timing observations, Planck sky maps
Numerical Methods	Signal processing, uncertainty estimation, regularization, inverse problems, quantitative modeling
Machine Learning	Python, scikit-learn, PyTorch, PLS/PCR, deep learning, predictive modeling
Scientific Tools	NumPy, pandas, SciPy, Jupyter, Git, Linux, reproducible analysis
Applied Engineering	RAG workflows, FastAPI, FAISS, LangGraph, SUMO/TraCI simulation

Selected Peer-Reviewed Publications

- El Byad, H., et al. (2025). “A NuSTAR Study of Quasiperiodic Oscillations from the Ultraluminous X-Ray Sources in M82.” Astrophysical Journal, 989(2), 202. First author. DOI
- Lai, E. V., et al. (2026). “High performance Stingray.” Astronomy and Computing, Article 101096. Co-author. DOI
- Bachetti, M., et al. (2023). “The High Energy X-ray Probe (HEX-P): Studying Extreme Accretion with Ultraluminous X-Ray Sources.” Frontiers in Astronomy and Space Sciences, 10, 1289432. Co-author. DOI

Education

Ph.D. in Mathematics & Computer Science **2022 – 2025**
University of Cagliari, Italy

M.Sc. in Applied Mathematics **2019 – 2021**
University of Cagliari, Italy

B.Sc. in Applied Mathematics **2015 – 2018**
Mohammed V University, Morocco

Selected Conference Presentations

- Oral - “Studying the aperiodic X-ray variability in the M82 Galaxy,” CNOC XII, Cefalù, 2022. Programme
- Poster - “QPOs in the M82's Ultraluminous X-ray sources,” FERO-10 (2022).
- Poster - “The aperiodic X-ray variability in the M82 Galaxy,” CNOC XIII, Alghero, 2024. Programme
- Poster - XMM-Newton 2024 Science Workshop, Madrid (2024).

Advanced Doctoral Training

- 4th Italian Astrostatistics School - Milan (2023).
- Hausdorff School: Inverse Problems for Multi-Scale Models - Bonn (2022).

Selected Certifications

- Machine Learning Foundations - University of Washington (Coursera, 2023).
- Fundamentals of Quantitative Modeling - University of Pennsylvania (Coursera, 2025).

Languages

English | Italian | French